Week 04 — Parametric Estimation: Yule–Walker & Least Squares

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The big picture. AR/ARMA parameters are estimated by method-of-moments (Yule–Walker), by least squares (forward/backward), or by recursively reconstructing the innovations.

We have modelled $\{X_t\}$ as an ARMA process; now we *fit* it. Given data $X_1, \dots, X_n$, estimate the AR coefficients $\phi_1, \dots, \phi_p$, the MA coefficients $\theta_1, \dots, \theta_q$, and the innovation variance σ_ε^2. Throughout $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$ is Gaussian white noise and the AR part is **causal** (roots of $\Phi(z)$ outside the unit circle), so that ε_t is uncorrelated with the past X_{t-k}, $k > 0$ — the one fact that makes everything work.

The two big ideas

There are two routes. **(1) Yule–Walker** = match moments: turn the AR recursion into a linear system in the autocovariances and solve once. **(2) Least squares** = minimise the squared one-step prediction errors $\sum \varepsilon_t^2$. For pure AR these coincide asymptotically; for ARMA, least squares is the only one that generalises, but the innovations must first be rebuilt.

Key definitions

Definition — Yule–Walker estimators (AR(p))

For a mean-zero causal AR(p), form the sample ACVS (mean known to be 0)

$$\hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} X_t X_{t+|\tau|},$$

collect $\hat{\boldsymbol{\gamma}}_p = (\hat{\gamma}_1, \dots, \hat{\gamma}_p)^\mathsf{T}$ and the symmetric Toeplitz matrix $\hat{\Gamma}_p = (\hat{\gamma}_{i-j})_{i,j=1}^p$ (diagonal $\hat{\gamma}_0$). Then

$$\hat{\boldsymbol{\phi}}_p = \hat{\Gamma}_p^{-1} \hat{\boldsymbol{\gamma}}_p, \qquad \hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0 - \sum_{j=1}^p \hat{\phi}_j \, \hat{\gamma}_j.$$

This is the empirical analogue of the population identities $\boldsymbol{\gamma} = \Gamma \boldsymbol{\phi}$ and $\gamma_0 = \sum_j \phi_j \gamma_j + \sigma_\varepsilon^2$.

Definition — Forward least-squares estimator (AR(p))

Write the AR equations for the observable times $t = p+1, \dots, n$ as $\mathbf{X}_F = F\boldsymbol{\phi} + \boldsymbol{\varepsilon}$, where $\mathbf{X}_F = (X_{p+1}, \dots, X_n)^\mathsf{T}$ and the rows of F are the lagged values $(X_{t-1}, \dots, X_{t-p})$. The forward LS estimator minimises

$$\mathrm{SS}_F(\boldsymbol{\phi}) = \|\mathbf{X}_F - F\boldsymbol{\phi}\|^2 = \sum_{t=p+1}^n \left(X_t - \textstyle\sum_{k=1}^p \phi_k X_{t-k} \right)^2,$$

giving the OLS solution $\hat{\boldsymbol{\phi}}_F = (F^\mathsf{T} F)^{-1} F^\mathsf{T} \mathbf{X}_F$ and $\hat{\sigma}_F^2 = \left\| \mathbf{X}_F - F\hat{\boldsymbol{\phi}}_F \right\|^2 / (n - 2p)$. The first p observations are discarded as conditioning values.

Definition — Backward & forward–backward LS (AR(p))

By *time reversibility*, regress the *first* $n-p$ values on their *future* lags: $X_t = \sum_{j=1}^{p} \phi_j X_{t+j} + \nu_t$, $t = 1, \dots, n-p$, i.e. $\mathbf{X}_B = B\boldsymbol{\phi} + \boldsymbol{\nu}$, giving $\hat{\boldsymbol{\phi}}_B = (B^\top B)^{-1} B^\top \mathbf{X}_B$ with the same $\boldsymbol{\phi}$. The **forward–backward** estimator $\hat{\boldsymbol{\phi}}_{FB}$ minimises the combined criterion $\mathrm{SS}_F(\boldsymbol{\phi}) + \mathrm{SS}_B(\boldsymbol{\phi})$ (stack F, B and $\mathbf{X}_F, \mathbf{X}_B$, then OLS). Simulations show it usually beats both.

Definition — Least squares for ARMA via residual reconstruction

For $\Phi(B)X_t = \Theta(B)\varepsilon_t$ the innovations are *unobserved*. LS chooses parameters minimising $S(\boldsymbol{\phi}, \boldsymbol{\theta}) = \sum_t \hat{\varepsilon}_t^2$, where for each candidate value the residuals are **reconstructed recursively** from the model equation after setting the q pre-sample innovations (and any pre-sample data) to zero, e.g. $\varepsilon_0 = \varepsilon_{-1} = \cdots = 0$. Nonlinear in general; closed form only in the pure-AR case (then it is forward LS).

Key results & formulas

Proposition — Yule–Walker equations

For a mean-zero causal AR(p) with ACVS γ_τ, for every $k \geq 1$

$$\gamma_k = \sum_{j=1}^{p} \phi_j \, \gamma_{k-j}, \qquad \text{i.e.} \qquad \boldsymbol{\gamma} = \Gamma \boldsymbol{\phi}, \quad \gamma_0 = \sum_{j=1}^{p} \phi_j \gamma_j + \sigma_\varepsilon^2.$$

Idea: multiply $X_t = \sum_j \phi_j X_{t-j} + \varepsilon_t$ by X_{t-k} and take expectations; causality kills $\mathbb{E}[\varepsilon_t X_{t-k}] = 0$ for $k > 0$. Replacing γ by $\hat{\gamma}$ gives the YW estimators.

Why causality is the whole trick

"Causal $\Rightarrow \mathbb{E}[\varepsilon_t X_{t-k}] = 0$" is the heart of it. X_{t-k} is built from innovations *strictly before* ε_t, and white noise is uncorrelated, so the cross term vanishes — leaving a clean linear system in the autocovariances. Without causality the noise leaks into the past and YW fails.

Proposition — YW = best linear predictor

The $\phi_1, \dots, \phi_k$ minimising the one-step MSE $\mathbb{E}\big[(Y_t - \sum_{i=1}^{k} \phi_i Y_{t-i})^2\big]$ satisfy exactly the Yule–Walker equations $\gamma_j = \sum_i \phi_i \gamma_{j-i}$. *Idea:* the objective is a positive quadratic $\gamma_0 - 2\sum_i \phi_i \gamma_i + \sum_{i,j} \phi_i \phi_j \gamma_{i-j}$; setting $\partial/\partial\phi_i = 0$ gives YW. So the AR coefficients *are* the optimal linear-prediction weights — which is why YW and LS agree asymptotically.

Proposition — time reversibility of a Gaussian AR

If $\{X_t\}$ is a stationary Gaussian AR(p), then $Y_t = X_{-t}$ is an AR(p) with the *same* coefficients and innovation variance. Equivalently $X_t = \sum_{j=1}^{p} \phi_j X_{t+j} + \nu_t$. *Idea:* a Gaussian process is determined by its ACVS, and $\gamma_{-\tau} = \gamma_\tau$; forward and reversed series have identical second-order structure. This is exactly what *legitimises* the backward regression.

Levinson–Durbin & a caveat

Levinson–Durbin: the YW system $\hat{\Gamma}_p \hat{\boldsymbol{\phi}}_p = \hat{\boldsymbol{\gamma}}_p$ is symmetric Toeplitz; this structure is exploited to solve it recursively in $O(p^2)$ instead of $O(p^3)$, producing as a by-product the PACF $\hat{\alpha}_k = \hat{\phi}_{k,k}$ and the error variance at each order.
Caveat: the LS estimators $\hat{\boldsymbol{\phi}}_F, \hat{\boldsymbol{\phi}}_B, \hat{\boldsymbol{\phi}}_{FB}$ are *not* constrained to be stationary ($\hat{\Phi}(z)$ may have roots inside the unit circle) — a problem for prediction, but fine for spectral estimation. Yule–Walker, by contrast, *always* yields a stationary fit.

Recipe — recursive innovation reconstruction (MA/ARMA). Solve the model equation for

ε_t, fix the q pre-sample innovations (and data) to zero, recurse forward, then minimise $S = \sum_t \hat{\varepsilon}_t^2$:

$$\text{MA(1):} \quad \varepsilon_t = X_t + \theta\varepsilon_{t-1},$$
$$\text{ARMA(1,1):} \quad \varepsilon_t = X_t - \phi X_{t-1} + \theta\varepsilon_{t-1} \quad (X_0 = \varepsilon_0 = 0).$$

Each $\hat{\varepsilon}_t$ is an explicit function of the parameters; because of the $\theta\hat{\varepsilon}_{t-1}$ feedback, S is generally *not* quadratic and must be minimised numerically.

Yule–Walker vs. least squares, in practice

For large n both give nearly the same AR estimates (e.g. $\hat{\phi} \approx \hat{\rho}_1$ for AR(1), which is also what method-of-moments gives). The differences are at the margins: YW is one linear solve and always stationary; LS extends cleanly to ARMA and to including a mean, but can stray outside the stationary region.

A tiny worked example

AR(1) Yule–Walker, by hand

A length-200 series gives $\hat{\gamma}_0 = 4.0$, $\hat{\gamma}_1 = 2.4$. For $p = 1$ the YW system is scalar, $\hat{\gamma}_1 = \hat{\phi}\,\hat{\gamma}_0$, so

$$\hat{\phi} = \frac{\hat{\gamma}_1}{\hat{\gamma}_0} = \frac{2.4}{4.0} = 0.6 = \hat{\rho}_1, \qquad \hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0 - \hat{\phi}\,\hat{\gamma}_1 = 4.0 - 0.6(2.4) = 2.56.$$

Check: $\hat{\gamma}_0(1 - \hat{\phi}^2) = 4.0(0.64) = 2.56$ matches $\gamma_0 = \sigma_\varepsilon^2/(1 - \phi^2)$. Stationarity: $\Phi(z) = 1 - 0.6z$ has root $z = 1.67$, modulus > 1, so the fit is causal.

Key takeaways

Key takeaways — the week's reflexes

- **Fit AR(p) by YW:** estimate $\hat{\gamma}_0, \ldots, \hat{\gamma}_p$, build the Toeplitz $\hat{\Gamma}_p$ and vector $\hat{\boldsymbol{\gamma}}_p$, solve $\hat{\boldsymbol{\phi}}_p = \hat{\Gamma}_p^{-1}\hat{\boldsymbol{\gamma}}_p$, then $\hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0 - \sum_j \hat{\phi}_j\hat{\gamma}_j$.
- **ACF shortcut:** dividing the whole system by $\hat{\gamma}_0$ replaces covariances by correlations, $\hat{\boldsymbol{\phi}}_p = \bar{\Gamma}_p^{-1}\hat{\boldsymbol{\rho}}_p$ ($\hat{\gamma}_0$ cancels; needed only for $\hat{\sigma}_\varepsilon^2$).
- **LS is regression:** forward/backward/forward–backward are all OLS $\hat{\boldsymbol{\phi}} = (A^\top A)^{-1}A^\top\mathbf{X}$. Remember $\hat{\sigma}^2$ divides by $n - 2p$, not n.
- **MA/ARMA:** the innovations are latent — *always* reconstruct them recursively (initial ε's set to 0) before squaring; expect a numerical minimisation.
- **Reflexes:** YW $\Rightarrow$ always stationary; LS $\Rightarrow$ flexible (ARMA, mean) but may be non-stationary; don't put γ_0 in the RHS vector.